

Definitions and Scope of Economics

2. Definitions and Scope of Economics; Goods, Wants, Wealth and Income

2.1 Why CST Students Should Study Economics

Economics is useful for CST students because the digital world is full of economic decisions. Technology products are developed, priced, distributed, purchased, regulated and evaluated through economic choices. Even when a product appears to be free, resources are being used somewhere: programmer time, servers, electricity, cloud storage, data, cybersecurity systems and user attention.

Examples from CST life and career:

- A student must decide whether to buy a laptop, a smartphone, a paid course or save money for future study.
- A software company must decide whether to offer a free version, a paid version or a subscription model.
- A startup must decide whether to spend money on programmers, servers, advertising or cybersecurity.
- An app developer must understand user demand and the value users receive from the app.
- A government must decide how much to invest in digital infrastructure, cybersecurity, ICT education and online public services.

All these situations involve choice, scarcity, cost, and benefit. These are central ideas in economics.

Core Message

Economics begins when resources are limited and choices must be made. CST students face scarcity of time, money, computing resources, internet bandwidth, attention, skilled labor and data.

2.2 Meaning and Definitions of Economics

Economics is commonly defined as the study of how individuals, firms, governments and societies make choices under scarcity. Human wants are unlimited, but resources are limited. Therefore, every society must decide what to produce, how to produce and for whom to produce. Every choice involves a cost, usually called opportunity cost.

2.1 Wealth-Based Definition

Early economists such as Adam Smith emphasized wealth. According to this view: Economics is the study of the nature and causes of the wealth of nations. This definition focuses on production, exchange, and accumulation of wealth.

Limitations

This definition is narrow because economics is not only about wealth. It is also about human welfare, choice, scarcity, employment, poverty, inequality, development, technology, and the environment.

2.2 Welfare-Based Definition

Alfred Marshall defined economics as a study of mankind in the ordinary business of life. According to this view: Economics studies how people earn income and how they use income to improve their welfare. This definition shifted attention from wealth to human welfare.

Example

A person earns money through work and spends it on food, housing, education, healthcare, and digital services. These choices affect welfare.

2.3 Scarcity-Based Definition

Lionel Robbins gave one of the most widely used definitions of economics. According to Robbins: Economics is the science which studies human behavior as a relationship between ends and scarce means which have alternative uses. This definition contains three important ideas:

Human wants are unlimited.

Resources are scarce.

Resources have alternative uses.

Example:

A student has Tk. 80,000. They may use it to buy:

A laptop,

A desktop computer,

A smartphone,

Online courses,

Programming books,

Software subscriptions.

The money is limited, but the possible uses are many. Therefore, the student must choose.

2.4 Growth and Development-Based Definition

Modern economics also studies growth and development.

It asks questions such as:

How can a country increase national income?

How can poverty be reduced?

How can employment be created?

How can technology increase productivity?

How can digital transformation improve the economy?

For CST students, this is important because technology is now one of the main drivers of productivity, innovation, and development.

Approach	Main Idea	Explanation	Limitations/Comments
Wealth definition	Economics studies the nature and causes of wealth.	Early economists such as Adam Smith emphasized production, exchange and accumulation of wealth.	Too narrow because economics is not only about wealth; it also studies welfare, scarcity, choice, employment, poverty, technology and development.
Welfare definition	Economics studies human welfare in the ordinary business of life.	Alfred Marshall shifted attention from wealth to human well-being. People earn income and use it to improve welfare.	Welfare is important, but the definition does not fully explain scarcity and alternative uses of resources.
Scarcity definition	Economics studies human behavior as a relationship between unlimited ends and scarce means having alternative uses.	Lionel Robbins emphasized scarcity, choice and opportunity cost. This is very useful for explaining everyday decisions.	It is analytically strong, but it may appear too neutral because it does not directly emphasize welfare, equity and development.
Growth and development view	Economics studies how societies increase output, reduce poverty and improve living standards.	Modern economics studies national income, employment, productivity, human capital, digital transformation and inclusive development.	Very relevant for developing countries and technology-driven economies.

For students, the scarcity definition is especially practical. For example, a student has limited time. If the student spends three hours watching entertainment videos, they cannot spend the same three hours learning Python, Java, networking or data analysis. The opportunity cost is the value of the best alternative use of time.

2.3 Definitions of Basic Economic Terms

The following terms are essential for understanding economics. They should be explained slowly with examples.

Term	Definition	Example
Scarcity	A situation where resources are limited compared to human wants.	Limited income, limited time, limited land, limited skilled workers, limited internet bandwidth, limited computer lab seats, limited cloud storage or limited study time.
Choice	Selecting one option from available alternatives.	Choosing whether to learn web development, app development, data science or cybersecurity first.
Opportunity cost	The value of the next best alternative sacrificed when a choice is made.	If a student buys a gaming phone instead of a laptop, the sacrificed laptop-related learning opportunity is part of the opportunity cost.
Production	Creating goods and services that satisfy wants.	Writing code, developing a mobile app, building a website, providing IT support.
Consumption	Using goods and services to satisfy wants.	Using mobile data, watching an online tutorial, using a software subscription.
Exchange	Giving something to receive something else.	Paying a subscription fee for cloud storage or exchanging programming labor for salary.
Distribution	Sharing income or output among people or factors of production.	A software firm pays salaries to developers, rent to office owners and profit to owners or investors.
Market	Any arrangement where buyers and sellers interact.	Daraz, Upwork, Google Play Store, app marketplaces or a local computer shop.
Price	The money value of a good or service.	The monthly price of an internet package or software license.
Incentive	A reward or penalty that influences behavior.	Students may learn programming faster if it improves job prospects or freelancing income.

2.4 Scope of Economics

The scope of economics means the subject matter covered by economics. Economics studies both individual choices and economy-wide issues. It also studies traditional sectors such as agriculture and industry, and modern sectors such as information technology, digital finance, artificial intelligence and platform-based services.

Microeconomics

Microeconomics studies individual economic units such as consumers, firms and markets. It explains demand, supply, price determination, consumer choice, production decisions and market competition. Example: A software company deciding the price of a mobile app is a microeconomic issue.

Macroeconomics

Macroeconomics studies the economy as a whole. It explains national income, inflation, unemployment, economic growth, government budget, money supply and international trade. Example: The effect of inflation on laptop prices or internet subscription costs is a macroeconomic issue.

Positive economics

Positive economics explains facts and cause-effect relationships. It describes what is, what was or what will be. Example: Higher internet cost reduces access to online learning. This statement can be tested with data.

Normative economics

Normative economics explains what ought to be. It involves value judgments. Example: The government should provide subsidized internet to poor students. This is a policy opinion and cannot be tested in the same way as a positive statement.

Traditional Areas of Economics	Modern and CST-Relevant Areas
Consumption, production, exchange and distribution	Digital consumption, software production, platform exchange and digital labor markets
Public finance and taxation	Taxation of digital services, e-commerce and online income
Money and banking	Mobile banking, fintech, payment gateways and cybersecurity
International trade	Global outsourcing, freelancing, software exports and digital services trade

Traditional Areas of Economics	Modern and CST-Relevant Areas
Economic development	ICT-led development, digital inclusion, skills, human capital and innovation
Welfare and inequality	Digital divide, access to devices, internet inequality and AI access gaps

3. Goods: Meaning and Classifications

3.1 Meaning of Goods

Goods are things that satisfy human wants. In everyday language goods often mean physical objects, but in economics goods may include services and digital products as well. Goods may be tangible, such as rice, clothes and computers, or intangible, such as software, online classes, cloud storage and technical support.

A good becomes economically important when it has utility and is scarce. For example, a mobile app satisfies wants, and programmer time is required to develop it. Therefore, even though it is digital, it is an economic good.

5. Goods

5.1 Meaning of Goods

Goods are things that satisfy human wants.

Goods may be physical or non-physical.

Examples: Rice, Clothes, Books, Computers, Mobile phones, Software, Internet services, Cloud storage, Online courses.

In economics, both goods and services are often included when discussing economic goods.

5.2 Classification of Goods

Goods can be classified in several ways.

5.2.1 Free Goods and Economic Goods

Free Goods

Free goods are available in nature without cost and are not scarce under normal conditions.

Examples: Sunlight, Air, Rainwater in some places.

Free goods usually do not have a market price.

However, some free goods can become scarce due to pollution or overuse. For example, clean air is becoming scarce in many cities.

Economic Goods Economic goods are scarce and have a price.

Examples: Food, Land, Books, Computers, Internet packages, Software licenses, Electricity.

Most goods studied in economics are economic goods.

5.2.2 Consumer Goods and Producer Goods

Consumer Goods Consumer goods are used directly by consumers to satisfy wants.

Examples: Food, Clothes, Mobile phones, Laptops used by students, Streaming subscriptions.

Producer Goods Producer goods are used to produce other goods and services.

Examples: Machinery, Factory equipment, Office computers, Servers, Software development tools, Cloud infrastructure.

A laptop used by a student for entertainment may be a consumer good. But the same laptop used by a programmer to develop software may be considered a producer good.

5.2.3 Durable Goods and Non-Durable Goods

Durable Goods Durable goods can be used for a long time.

Examples: Computer, Car, Furniture, Smartphone, Printer.

Non-Durable Goods Non-durable goods are consumed quickly or used only once.

Examples: Food, Fuel, Paper, Ink, Electricity used in one billing period.

5.2.4 Intermediate Goods and Final Goods

Intermediate Goods Intermediate goods are used as inputs to produce final goods.

Examples: Raw materials, Computer chips used in laptop manufacturing, Source code modules used in software production, APIs used to build applications.

Final Goods Final goods are ready for final use by consumers or firms.

Examples: A completed mobile app, A finished laptop, A packaged software product, A final website delivered to a client.

5.2.5 Private Goods and Public Goods

Private Goods Private goods are goods from which people can be excluded if they do not pay. One person's consumption also reduces availability for others.

Examples: Food, Clothes, Private tuition, Paid software subscription.

Public Goods Public goods are non-excludable and non-rival. This means: It is difficult to stop people from using them. One person's use does not reduce another person's use.

Examples: National defense, Street lighting, Public safety, Basic public knowledge, Open-source knowledge in some cases.

For CST students, open-source software has some public-good-like characteristics, although it may also have private elements depending on licensing and usage.

5.2.6 Substitute Goods and Complementary Goods

Substitute Goods Substitute goods can be used in place of each other.

Examples: Tea and coffee, Desktop and laptop, Android and iPhone, Zoom and Google Meet, Windows and Linux for some users.

Complementary Goods Complementary goods are used together.

Examples: Printer and ink, Computer and software, Smartphone and internet package, Laptop and charger, Cloud service and internet connection.

5.2.7 Normal Goods and Inferior Goods

Normal Goods Demand for normal goods increases when income increases.

Examples: Better-quality clothes, Better smartphones, Higher-speed internet, Paid online courses.

Inferior Goods Demand for inferior goods may decrease when income increases.

Examples: Low-quality food items, Second-hand devices, Very cheap substitutes.

The word "inferior" does not always mean bad. It only means that people buy less of it when their income rises.

5.2.8 Necessities, Comforts, and Luxuries

Necessities Necessities are essential for life and basic functioning.

Examples: Food, Clothing, Shelter, Basic education, Basic healthcare, Basic internet access for students in modern education.

Comforts Comforts make life easier and more pleasant.

Examples: A good chair, A better laptop, Air conditioning, High-speed internet.

Luxuries Luxuries are expensive goods that provide extra satisfaction beyond basic needs.

Examples: Luxury cars, High-end gaming computers, Premium smartphones, Expensive watches, Luxury apartments.

5.2.9 Digital Goods

Digital goods are goods that exist in digital form.

Examples: Software, E-books, Online courses, Digital music, Video games, Mobile apps, Cloud storage, AI tools.

Digital goods have some special features:

They can be copied at very low cost.

They can be distributed globally through the internet.

Their value may depend on users, networks, and data.

They may be sold through subscription models.

They may have almost zero marginal cost after development.

Example: The first copy of software may require huge investment. But after it is developed, selling one more copy may cost very little.

3.2 Classification of Goods

Classification	Meaning	Examples	Note
Free goods	Goods available without cost and not scarce under normal conditions.	Air and sunlight.	Clean air may become scarce due to pollution; free online resources may still require servers and data.
Economic goods	Goods that are scarce and usually have a price.	Food, books, computers, internet packages.	Most software, data services, cloud storage and paid apps are economic goods.
Consumer goods	Goods used directly by consumers to satisfy wants.	Food, clothes, smartphones, entertainment subscriptions.	A student using a laptop for study or entertainment treats it as a consumer good.
Producer goods	Goods used to produce other goods and services.	Machinery, office computers, servers, development tools.	The same laptop used by a programmer to earn income becomes a producer good.
Durable goods	Goods that can be used for a long period.	Computer, phone, furniture, printer.	Hardware and networking devices are durable goods.
Non-durable goods	Goods consumed quickly.	Food, fuel, electricity, paper.	Monthly mobile data or electricity used for computing is non-durable.
Intermediate goods	Goods used as inputs to produce final goods.	Raw materials, chips, APIs, code libraries.	A code library used to build an app is an intermediate input.
Final goods	Goods ready for final use.	A finished laptop, completed app or delivered website.	A final software product delivered to a client is a final good.
Private goods	Goods where non-payers can be excluded and one person's use reduces availability.	Food, paid software, private tutoring.	A paid software license is normally a private good.
Public goods	Goods that are non-excludable and non-rival.	National defense, street lighting.	Open-source knowledge may have public-good-like features, though licenses and services can create private elements.
Substitute goods	Goods that can be used in place of each other.	Tea and coffee; laptop and desktop; Zoom and Google Meet.	Users may switch from one programming language, platform or software to another.
Complementary goods	Goods used together.	Printer and ink; laptop and charger; smartphone and internet.	Computer hardware, operating systems, software and internet are often complements.
Normal goods	Goods whose demand rises as income rises.	Better phones, better internet packages, paid courses.	Students may buy better devices after income rises.
Inferior goods	Goods whose demand may fall as income rises.	Very low-quality goods or second-hand devices in some cases.	Inferior does not mean morally bad; it describes demand behavior.
Digital goods	Goods existing in digital form.	Software, e-books, online courses, games, cloud storage.	Digital goods often have high first-copy cost and low marginal cost of distribution.

Important Point on Digital Goods

Digital goods are special because after the first version is developed, the cost of supplying one more copy may be very low. However, they are not costless. They require developers, servers, maintenance, updates, cybersecurity, electricity, marketing and customer support.

4. Wants: Meaning, Features and Classifications

4.1 Meaning of Wants

Wants are desires for goods and services that can provide satisfaction. Human wants are the starting point of economic activity. People work, earn, save, invest, produce and exchange because they have wants. Some wants are basic, such as food and shelter. Some wants are related to comfort, status, entertainment, learning or career development.

Examples of wants include a laptop, internet access, programming skill, software tools, online courses, internship opportunities, freelancing income, better lab facilities and access to AI tools.

4.2 Features of Wants

Feature	Explanation	Example
Wants are unlimited	People always want more or better goods and services.	After buying a smartphone, a student may want a better laptop, faster internet and paid courses.
A particular want is satiable	One specific want can be satisfied at a given time.	Hunger can be satisfied by eating food, although it returns later.
Wants are recurring	Many wants come back again and again.	Food, water, electricity, mobile data and internet subscriptions are recurring wants.
Wants are complementary	Satisfaction of one want may create another related want.	Buying a computer creates wants for software, internet, antivirus, electricity and maintenance.
Wants change over time	Age, income, technology and society change wants.	Online learning and AI tools are now more important than they were in the past.
Wants differ among people	Different people value different goods.	A gamer, programmer and graphic designer may want different computer specifications.
Wants are influenced by advertising	Media, influencers and algorithms can shape wants.	A student may want a device after seeing online reviews or social media advertisements.

4.3 Classifications of Wants

Type of Want	Meaning	Examples
Necessaries	Wants that must be satisfied for survival or basic functioning.	Food, shelter, basic clothing, basic healthcare, education; for many students, basic internet access and a digital device.
Comforts	Wants that make life easier and improve efficiency.	Comfortable chair, better internet speed, air-conditioned classroom, good study environment.
Luxuries	Wants that provide extra pleasure, prestige or status.	Luxury car, very expensive phone, high-end gaming computer.
Individual wants	Wants of a single person.	A student wants a laptop; a programmer wants a mechanical keyboard.
Collective wants	Wants of a group or society.	Roads, public universities, public Wi-Fi, national cybersecurity and public health systems.
Present wants	Wants needing immediate satisfaction.	A student needs internet today to submit an assignment.
Future wants	Wants expected in the future.	A student saves money to buy a laptop next semester.

5. Wealth and Income

5.1 Wealth

In economics, wealth means valuable assets that can satisfy wants and have exchange value. Wealth can be physical, financial or digital. Examples include land, houses, machinery, computers, vehicles, money, shares, bonds, software licenses, patents, databases, domain names and business assets.

For something to be considered wealth in the economic sense, it usually needs four characteristics:

- Utility: it must satisfy human wants.
- Scarcity: it must be limited in supply.
- Transferability: it should be transferable from one person or organization to another.
- Market value: it should normally have exchange value or sale value.

Human qualities such as honesty, intelligence and skill are valuable, but they are not always called wealth in the narrow traditional sense because they cannot be directly transferred. However, modern economics recognizes human capital as a very important source of productivity and income.

Type of Wealth	Meaning	Examples
Personal wealth	Assets owned by an individual.	Savings, house, laptop, land, vehicle.
Social wealth	Assets owned or used collectively by society.	Roads, bridges, public universities, public hospitals, public data infrastructure.
National wealth	All valuable assets of a country.	Natural resources, infrastructure, industries, universities, human capital, ICT infrastructure.
Digital wealth	Valuable digital assets and systems.	Software, data, algorithms, patents, digital platforms, cloud systems, domain names.

5.2 Income

Income is the flow of money or benefits received by a person, firm or country over a period of time. It is different from wealth because wealth is a stock, while income is a flow. A person may own a house worth Tk. 50 lakh; this is wealth. If the person rents the house and receives Tk. 30,000 per month, that monthly rent is income.

Type of Income	Meaning	Example
Money income	Income measured in money terms.	A software engineer earns Tk. 80,000 per month.
Real income	The amount of goods and services that money income can buy.	If salary rises but prices rise faster, real income may fall.
Personal income	Total income received by an individual from all sources.	Salary, rent, interest, freelancing income and business profit.
Disposable income	Income left after paying personal taxes.	Disposable income = personal income - personal taxes.
National income	Total income earned by the people of a country in a year.	Wages, rent, interest and profit earned in the whole economy.

Basis	Wealth	Income
Nature	Stock concept	Flow concept
Time reference	Measured at a point in time	Measured over a period of time
Example	A laptop, land or bank balance	Salary, freelancing earnings or rent per month
CST example	A server, software license or database	Income from software job, app sales or online freelancing

Recap

Economics studies choice under scarcity. Goods satisfy wants. Wants are unlimited and change over time. Wealth is a stock of valuable assets, while income is a flow of earnings over time. CST students should understand these ideas because technology decisions also involve cost, benefit, scarcity and choice.

6. Money, Near Money, Credit Money and Utility

6.1 Money: Meaning and Evolution

Money is anything that is generally accepted as a medium of exchange for goods and services and for the repayment of debt. The most important word in this definition is generally accepted. If people do not accept something in exchange, it cannot function as money.

Money developed because the barter system was inefficient. In a barter system, goods are directly exchanged for goods. For example, a farmer may exchange rice for clothes. This system becomes difficult when wants do not match.

Problem of Barter	Explanation	Simple Example
Double coincidence of wants	Both parties must want what the other has.	A programmer wants rice, but the farmer may not need software.
Lack of common measure of value	It is difficult to compare values of different goods.	How many kg of rice equals one software service?
Indivisibility of goods	Some goods cannot be divided easily.	A cow cannot be divided to buy small items.
Difficulty of storing value	Some goods lose value over time.	Fish, fruits or vegetables cannot store value for long.
Difficulty of deferred payment	Future payments are hard to specify.	A loan in terms of rice may be unfair if rice quality or price changes.

The evolution of money can be explained in stages: barter, commodity money, metallic money, paper money, bank money and digital money. Modern students mostly experience money through cash, bank accounts, mobile financial services, debit cards, online transfers and digital wallets.

Stage	Form of Money	Example
Commodity money	Useful goods used as money.	Cattle, salt, rice, gold.
Metallic money	Coins made from metals.	Gold, silver or copper coins.
Paper money	Currency notes issued by monetary authority.	Bangladeshi taka notes.
Bank money	Deposits transferable through banking system.	Cheques, debit cards, bank transfers.
Digital money	Electronically stored and transferred money.	Mobile banking balance, digital wallets, online payments.

6.2 Functions of Money

Function	Meaning	Example
Medium of exchange	Money is used to buy and sell goods and services.	A student pays Tk. 1,000 for an online course.
Measure of value	Money measures and compares the value of goods.	A laptop costs Tk. 75,000 and a phone costs Tk. 25,000.
Store of value	Money can be saved for future use.	A student saves money to buy a laptop next semester.
Standard of deferred payment	Money is used for future payments.	Loan installment, credit card bill or EMI payment.
Transfer of value	Money transfers purchasing power across persons or places.	Parents send money to a student through mobile banking.

Good money should have general acceptability, durability, portability, divisibility, uniformity, recognizability, stability of value and difficulty of counterfeiting. Digital money also requires secure authentication, encryption, reliable databases, fraud detection and system availability.

CST Connection: Digital Money

Digital money depends on computer networks, databases, cryptography, authentication, cybersecurity, user interface design and transaction processing. Therefore, CST students can connect money with software engineering, database management, information security and fintech.

7. Near Money

Near money means assets that are not directly used as money but can be quickly converted into money. Near money is highly liquid but not as liquid as cash or current account deposits. It is often safer than many long-term assets and may earn interest.

Examples of Near Money	Why It Is Near Money
Savings deposits	Can be withdrawn or transferred, but not always used directly for every payment.
Fixed deposits	Can be converted into money, sometimes after maturity or with penalty.
Treasury bills	Short-term government securities that can be sold or redeemed.
Government bonds	Can be sold in financial markets, though price may vary.
Savings certificates	Can be converted into cash according to rules.

Basis	Money	Near Money
Meaning	Directly accepted as a medium of exchange.	Easily convertible into money but not directly used for all payments.
Liquidity	Most liquid.	Less liquid than money but more liquid than land or machinery.
Use in payment	Can be used directly.	Usually must be converted first.
Examples	Currency, coins, current account deposits.	Fixed deposits, treasury bills, bonds, savings certificates.

8. Credit Money

Credit money is money created through credit or lending activities. It is based on trust and legal obligation. When banks give loans, they create deposits in the banking system. These deposits can be used for payments and therefore function as money. Example: A startup may take a bank loan to buy computers, rent an office, and hire programmers.

A simplified process of credit creation is as follows:

1. A person deposits Tk. 100,000 in a bank.
2. The bank keeps a portion as reserve, for example Tk. 10,000.
3. The bank lends the remaining Tk. 90,000 to a borrower.
4. The borrower spends the money, and the receiver deposits it in another bank.
5. The second bank again keeps a reserve and lends the rest.
6. Through repeated lending and depositing, the banking system creates more deposit money than the original cash deposit.

Importance of Credit Money	Risks of Credit Money
Supports business investment and expansion.	Excessive credit may increase inflation.
Helps consumers buy durable goods through installment or credit.	Bad loans can weaken banks and financial stability.
Promotes trade, production and employment.	Borrowers may fall into debt traps.
Helps startups purchase equipment and hire workers.	Weak regulation can encourage unsafe lending.
Makes modern payment systems more flexible.	Digital lending can create privacy, fraud and consumer protection issues.

Credit Money and Fintech

Credit scoring, digital lending, mobile banking, fraud detection and online payment systems depend on data, algorithms and secure software. CST students may work in systems that process financial data; therefore, they must understand both economic value and digital risk.

9. Utility

9.1 Meaning of Utility

Utility is the want-satisfying power of a good or service. It is the satisfaction, benefit or usefulness that a consumer expects to receive from consuming a good or service. A glass of water gives utility to a thirsty person. A laptop gives utility to a student by helping with learning, programming, communication and entertainment.

Utility in economics is subjective. It does not mean moral goodness or social usefulness. A thing may give utility to someone even if it is harmful. For example, cigarettes may give satisfaction to a smoker, although they are harmful to health.

Feature of Utility	Explanation	Example
Subjective	Utility differs from person to person.	A gaming laptop gives high utility to a gamer but lower utility to a casual user.
Depends on intensity of want	Utility is higher when the want is stronger.	Water has high utility to a thirsty person.
Not the same as usefulness	Economics focuses on satisfaction, not moral value.	A harmful product may still give utility to a consumer.
Changes with time and place	The same good may give different utility in different situations.	A power bank has high utility during electricity failure.
Diminishes with repeated consumption	Additional satisfaction from extra units usually falls.	The tenth tutorial video in one sitting may give less benefit than the first.
Difficult to measure exactly	Satisfaction is psychological and personal.	A student can rank choices but may not measure exact satisfaction numerically.

9.2 Types of Utility

Type	Meaning	Example
Total utility	Total satisfaction from consuming all units of a good.	Total satisfaction from watching four tutorial videos.
Marginal utility	Additional satisfaction from consuming one more unit.	Extra benefit from watching one more tutorial video.
Cardinal utility	Utility measured in numerical units.	First cup of tea gives 20 utils; second gives 15 utils.
Ordinal utility	Utility ranked by preference rather than exact measurement.	A student prefers laptop over tablet over smartphone.
Form utility	Utility created by changing the form of a good.	Raw code becomes useful software after design, testing and debugging.
Place utility	Utility created by making a good available at the right place.	Cloud storage allows access from any location.
Time utility	Utility created by making a good available at the right time.	A server backup is valuable when a system fails.
Possession utility	Utility created by transferring ownership or control.	Buying a software license gives legal access to software.
Service utility	Utility created by providing services.	IT support, cybersecurity service, teaching and software maintenance.

9.3 Law of Diminishing Marginal Utility

The law of diminishing marginal utility states that as a person consumes more and more units of the same good, keeping other things constant, the additional satisfaction from each extra unit tends to decrease. This law explains why people usually do not consume unlimited quantities of the same good even if they like it.

Formula: $\text{Marginal Utility} = \frac{\text{Change in Total Utility}}{\text{Change in Quantity Consumed}}$

Number of Tutorial Videos Watched	Total Utility	Marginal Utility	Interpretation
1	20	20	The first video gives high satisfaction because the student learns something new.
2	35	15	The second video is useful but gives less additional satisfaction.
3	45	10	Learning continues but at a lower additional rate.
4	50	5	The student may become tired; extra gain is small.
5	50	0	The fifth video adds no extra satisfaction.
6	45	-5	Too much watching may create confusion or fatigue.

In this example, total utility rises at first, then stops rising, and finally falls. Marginal utility falls continuously and becomes zero at the fifth video. When marginal utility becomes negative, total utility falls.

Explanation

Imagine watching programming tutorial videos for six hours continuously. At first, the tutorials are useful. Later, tiredness reduces concentration. This makes the law of diminishing marginal utility easy to understand.

10. CST-Focused Applications of Basic Economic Concepts

These examples help students see that economic concepts operate in the digital economy as strongly as in traditional markets.

Application	Economic Explanation	CST Example
Software as an economic good	Software satisfies wants, requires scarce resources to produce and has market value. Development needs programmers, designers, testers, servers and maintenance.	Paid apps, antivirus, accounting software, learning management systems.
Data as wealth	Data can be a valuable asset because it helps firms understand users, improve products, target services and train AI models.	Customer data, transaction data, health data, education data, search data.
Cloud computing and scarcity	Cloud services appear virtual but require physical servers, electricity, cooling systems, storage, bandwidth and skilled engineers.	Google Drive, AWS, Azure, institutional cloud services.
Digital platforms and wants	Platforms influence wants using recommendations, advertisements and social networks.	YouTube recommendations, Facebook ads, e-commerce suggestions.
AI tools and utility	AI tools provide utility by helping users write, code, translate, summarize, analyze and learn, but utility depends on accuracy and responsible use.	Chatbots, code assistants, image generators, data analysis tools.
Fintech and money	Digital money requires software, databases, authentication, encryption and cybersecurity.	Mobile banking, QR payments, payment gateways, digital wallets.
Opportunity cost in skills	Students must choose what to learn because time is limited.	Learning Python instead of Java, cybersecurity instead of app development, freelancing instead of exam preparation.
Digital divide	Access to devices, internet and paid tools affects educational and employment opportunities.	Students without laptops or stable internet face lower digital learning opportunities.